Ryan Schmelzle

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Summary

Applied computer vision researcher building reliable perception pipelines for real-world sensor data. Experience in 3D reconstruction, AI image analysis, and error detection, with an emphasis on deploying robust models for automated visual inspection.

Seeking roles in computer vision, automated visual inspection, and production-scale perception systems.

Experience

Graduate Researcher Jan. 2025 – Present
IRON Labs, UNC Chapel Hill

- Leading development of a robotic teleoperation and perception pipeline integrating computer vision, neural scene reconstruction, and image analysis.
- Designing and evaluating learning-based and geometric scene representations to support visual inspection, remote navigation, situational awareness, and error detection.
- Performing error detection and analyzing failure conditions in visual reconstructions under real-world constraints (e.g. sensor noise, occlusion, dynamic scenes).
- Investigating human-robot alignment between scene perception, detection outputs, and operator intent.

Graduate Researcher Apr. 2023 – Dec. 2024
Visual Computing & Augmented Intelligence Lab, UNC Chapel Hill

- Led development on prototype AR glasses to integrate computer vision and digital signal processing techniques.
- Designed a novel spatial audio filtering method and implemented it on a real-time audio-processing pipeline.
- Ongoing work involves in hardware design for visual localization integration and object recognition.

Undergraduate Researcher Aug. 2022 – May. 2023
Graphics and Virtual Reality Group, UNC Chapel Hill

- Contributed to neural rendering research for high-speed telepresence on sparse RGB-D camera setups under Henry Fuchs.
- Assisted with dataset preparation, baseline implementation, and experimental evaluation; co-authored an IEEE TVCG paper on few-camera neural telepresence systems.

Software Engineering Intern May. 2022 – Aug. 2022
Ford Motor Company (Remote), Dearborn MI

- Developed and maintained components for a data-driven vehicle recommendation platform for large-scale user interaction.
- Implemented backend services, frontend interfaces and database solutions using Java, JavaScript, SQL, and Vue.js.
- Worked within CI/CD pipelines and code-quality frameworks (Git, Jenkins, SonarQube).

Projects

Real-Time Robotic Teleoperation & Neural Scene Reconstruction *IRON Labs, UNC* - Ongoing

- Developing a vision-driven robotic teleoperation system supporting neural scene reconstruction and remote environment inspection deployed on a Boston Dynamics SPOT quadruped in simulated search-and-rescue scenarios.
- Building and maintaining **Vyse**, a real-time task-dispatch framework for coordinating distributed imaging workloads, data management, and external tooling.
- Designing a reconfigurable vision-analysis pipeline for quality assessment, error detection, anomaly recognition, and AI-based salience estimation across automated image-processing platforms.

Autoencoder-Based Neural Depth Estimation *Personal Project* - 2024

- Designed and trained an autoencoder-based monocular depth estimation pipeline using CNNs in TensorFlow/Keras.
- Created a pipeline for generating synthetic datasets using Blender with variable simulated conditions for training.

Education

University of North Carolina at Chapel Hill Chapel Hill, NC

Ph.D. Candidate, Computer Science Aug. 2023 - Expected May 2028
Focus: Computer vision, robotic perception, scene reconstruction

M.S. in Computer Science Aug. 2023 - Dec. 2024
Focus: Computational perception, audio-visual processing systems

B.S. in Computer Science & Communication Studies Aug. 2019 - May 2023
GPA: 3.92 / 4.0 Phi Beta Kappa; Honors Carolina; Highest Distinction

Relevant Courses: Computer Vision, Deep Learning, Computational Imaging, Computational Geometry, Neural Rendering

Skills

Core Proficiencies: Image processing, computer vision, computational geometry, 3D reconstruction, deep learning
Tools & Languages: Python, C++, SQL, OpenCV, ROS2, Blender, Git, Linux, PyTorch, Tensorflow